

# Grant application pre-proposal form 2025

## NWO Talent Programme – Veni scheme

Applied and Engineering Sciences  
Science  
Social Sciences and Humanities



*Version: June 2025*

Information about rules and guidelines (the 'explanatory notes') are embedded in this form. Please expand the explanatory notes for information about terms, conditions and requirements.

Note: The explanatory notes are automatically visible in Google Docs. To view them on Apple devices, select 'Outline' under the 'View' tab in Word.

Please **make sure the Explanatory notes are not visible in the PDF version of your pre-proposal**. To achieve this, delete the explanatory note text boxes before submitting your pre-proposal.

## 1. NWO domain and field of research

### 1a. NWO domain

Science (ENW) - Computer Science

### 1b. Main field of research

Indicate the main field of research and (if applicable) other fields of research, in order of relevance, by selecting names and codes from the dropdown menu. For more information, see the explanatory notes.

|                                                | Code/Field of research:                         |
|------------------------------------------------|-------------------------------------------------|
| Main field of research:                        | 16.90.00 Computer science, other                |
| Other field(s) of research<br>(if applicable): | 16.30.00 Theoretical computer science           |
|                                                | 16.40.00 Information systems, databases         |
|                                                | Choose fourth field of research (if applicable) |

## 2. Evidence-based curriculum vitae

### 2a. Academic profile

*For more information, expand the explanatory notes  
(Min. 400 – max. 700 words)*

#### *Academic Profile:*

I am an assistant professor of cybersecurity at TU Delft focusing on cryptography and privacy. My research addresses a central scientific and societal question:

*How can we safely use untrusted computing platforms while preserving our security and privacy?*

By combining rigorous theory with hands-on engineering, I design systems that achieve provable security and are applicable to the real-world.

#### **Academic Background and Achievements**

I hold a BSc (2016) and MEng (2018) in Electrical Engineering and Computer Science from MIT, with a minor in Mathematics. At MIT, I conducted research in distributed algorithms under Nancy Lynch. My work on leader election in wireless networks (**KO10**: Best Paper Award at SIROCCO, published in the journal of *Theoretical Computer Science*) was my first independent contribution as a Master student, demonstrating my ability to originate new ideas and bring them to impactful publication.

Motivated by the interplay between theory and practice, I pursued a PhD at Brown University (2018–2023) with Roberto Tamassia. I investigated encrypted outsourced databases, through a lens of both attacks and constructions. This work was recognized with a Best Student Paper Award, and resulted in three ACM CCS and a VLDB publications.

I also joined the **Bridging Privacy Definitions Working Group** (MIT, Brown, Harvard, University of Chicago, ETH Zurich, and others). Our joint paper (**KO7**), “*A principled approach to defining anonymization*,” connects formal privacy research with legal frameworks.

| Education                                                                                                                                                                           | Awards                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>MIT</b> Bachelor of Science 2016<br><b>MIT</b> Master of Engineering 2018<br><b>Brown University</b> Master of Science 2020<br><b>Brown University</b> Doctor of Philosophy 2023 | <b>Best Paper Award</b> in SIROCCO 2017<br><b>MIT Licklider UROP Award</b> 2017<br><b>Best Student Paper Award</b> in ISC 2019 |

#### **Transition to Independence**

During my PhD, I progressively took on leadership roles in research projects. In **KO1** (Best Student Paper Award), the only other author was my PhD advisor; I identified the attack vector, developed the analysis, and led the writing. In **KO2–KO3** and **KO5** (published at CCS and VLDB), I played a leading role within the team, contributing substantially to the protocol design, formal analysis, and technical writing, while also guiding students responsible for the implementation. In **KO4**, completed after defending my PhD, I independently drove the entire process - from protocol design and formal analysis to implementation - with only my advisor as co-author.

Since joining TU Delft, I have broadened my agenda to multiparty computation and secure data sharing. I pursued collaborations within the university with colleagues like George Smaragdakis, Kaitai Liang, and Zekeriya Erkin, as well as with the European Space Agency and Centrum Wiskunde & Informatica. The Veni will allow me to consolidate these efforts into a distinct and sustainable research program.

My track record spans theory, systems, and security venues (TCS, VLDB, CCS) and includes two best paper awards. I serve on **program committees** of security conferences (CCS, USENIX Security, PETS) and give invited talks, reflecting international recognition. I also co-authored a TU Delft feature article, “**Battling the Data Hunger of**

**Big Tech”,** making privacy research accessible to the wider public. This reflects my broader engagement and commitment to translating scientific work into societal impact. I maintain a strong international network across academia and industry.

I am committed to diversity and inclusion. At Brown, I contributed to the **Application Feedback Program for Underrepresented Applicants**. I have also participated in the **PETs Symposium Mentorship Event**, providing guidance to early-career researchers in privacy-enhancing technologies. At TU Delft, I host invited talks showcasing women in cybersecurity from both academia and government, and I participate in the **F+Cube (Future Female+ Faculty) mentorship program**, an award-winning initiative that supports underrepresented PhDs and postdocs in pursuing academic careers. These activities reflect my dedication to developing a stronger, more diverse academic community.

**Future Vision**

There is a pressing need for systems that allow the secure use of untrusted resources while safeguarding privacy. My Veni project will consolidate the independent line of research I have already begun, advancing searchable encryption, a technology already adopted in industry but whose leakage risks remain insufficiently understood. I will analyze leakage in multi-attribute databases under realistic adversaries and design defenses. The Veni will be the decisive step in establishing a sustainable and distinct research program under my sole leadership, completing my transition to full academic independence.

|                                                                  |     |
|------------------------------------------------------------------|-----|
| Word count 2a Academic profile<br>(min. 400 and max. 700 words): | 700 |
|------------------------------------------------------------------|-----|

## 2b. Key output

For more information, expand the explanatory notes.

Information about the culture and standards of the scientific field [optional in max. 50 words]

In theoretical computer science, including KO10, authorship is listed alphabetically, reflecting field standards rather than relative contribution. In computer security, the “Big Four” conferences are ACM CCS, USENIX Security, IEEE S&P, and NDSS. In databases, VLDB and SIGMOD hold similar status.

10 key outputs max. Total word limit: minimum 400 words, maximum 700 words, excl. references, URLs, output types and indicators.

### Key output 1

Open Access: No

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:          | Markatou, Evangelia Anna, and Roberto Tamassia. "Full database reconstruction with access and search pattern leakage." In International Conference on Information Security, pp. 25-43. Cham: Springer International Publishing, 2019.                                                                                                                                                                                                                                                                                               |
| URL:                | <a href="https://doi.org/10.1007/978-3-030-30215-3_2">https://doi.org/10.1007/978-3-030-30215-3_2</a>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Type:               | Conference paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Quality indicators: | 1) Academic awards, prizes and/or grants directly related to this output<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                                                         |
| Motivation:         | I demonstrated that seemingly harmless leakage, knowing whether two encrypted queries are the same, enables full reconstruction of one-dimensional encrypted databases. As first author, I identified the attack vector, developed the formal analysis, and led the writing. It received the Best Student Paper Award at ISC 2019, marking its novelty and impact. This work laid the foundation for my subsequent CCS, VLDB, PETS papers (KOs 2-6), which extended the results to multi-dimensional settings and protocol designs. |

### Key output 2

Open Access: No

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:          | Falzon, Francesca, Evangelia Anna Markatou, Akshima, David Cash, Adam Rivkin, Jesse Stern, and Roberto Tamassia. "Full database reconstruction in two dimensions." In Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security, pp. 443-460. 2020.                                                                                                                                                                                                                                                                                              |
| URL:                | <a href="https://doi.org/10.1145/3372297.3417275">https://doi.org/10.1145/3372297.3417275</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Type:               | Conference paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Quality indicators: | 1) Academic interest, other<br>2) Academic collaboration and/or interdisciplinary engagement<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Motivation:         | This was my first major external collaboration. We solved a five-year open problem: showing that leakage from two-dimensional encrypted databases enables full reconstruction. As first co-author, I led the attack design and formal analysis, and guided students in the experimental evaluation. This project marked an important step toward my independence, as I initiated and led key aspects of a collaboration beyond my home institution. The paper was published at ACM CCS, highlighting both the significance of the result and my leadership in collaboration. |

### Key output 3

Open Access: No

|                     |                                                                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:          | Markatou, Evangelia Anna, Francesca Falzon, Roberto Tamassia, and William Schor. "Reconstructing with less: Leakage abuse attacks in two dimensions." In Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security, pp. 2243-2261. 2021. |
| URL:                | <a href="https://doi.org/10.1145/3460120.3484552">https://doi.org/10.1145/3460120.3484552</a>                                                                                                                                                                        |
| Type:               | Conference paper                                                                                                                                                                                                                                                     |
| Quality indicators: | 1) Academic interest, other<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                       |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Motivation:</b> | Following KO2, this paper showed that database reconstruction in two-dimensions is possible even with limited leakage, lowering the attack threshold. As first co-author, I led the protocol design and formal analysis, and expanded my expertise by utilising mathematical tools from geometry and statistical estimation. The publication at ACM CCS highlights the significance of this contribution to the international security community. |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Key output 4

Open Access: Yes

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reference:</b>          | Markatou, Evangelia Anna, and Roberto Tamassia. "Reconstructing with Even Less: Amplifying Leakage and Drawing Graphs." In Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security, pp. 4777-4791. 2024.                                                                                                                                                                                                                                                                                    |
| <b>URL:</b>                | <a href="https://doi.org/10.1145/3658644.3670313">https://doi.org/10.1145/3658644.3670313</a>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Type:</b>               | Conference paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Quality indicators:</b> | 1) Academic interest, other<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Motivation:</b>         | This paper advanced the attack landscape by showing that databases of arbitrary dimension are also susceptible to reconstruction, whereas prior work focused on up to two dimensions. I was the first author, responsible for conception, analysis, implementation, and writing. With only my advisor as co-author, this project highlights my ability to independently drive research from idea to publication. Its acceptance at ACM CCS demonstrates the continuing academic and community interest in this line of work. |

#### Key output 5

Open Access: No

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reference:</b>          | Francesca Falzon, Evangelia Anna Markatou, Zachary Espiritu, and Roberto Tamassia. 2022. Range Search over Encrypted Multi-Attribute Data. Proc. VLDB Endow. 16, 4 (December 2022), 587–600.                                                                                                                                                                                                                                                                                                                                                     |
| <b>URL:</b>                | <a href="https://doi.org/10.14778/3574245.3574247">https://doi.org/10.14778/3574245.3574247</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Type:</b>               | Conference paper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Quality indicators:</b> | 1) Academic interest, other<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Motivation:</b>         | Complementing my attack papers, this work proposed new constructions for performing range queries on encrypted multi-attribute databases, balancing efficiency with security. I was first co-author, responsible for protocol design and supervising students in the implementation. The system was released as open source, demonstrating my commitment to transparency and practical impact. The publication at VLDB, a database venue, reflects the interdisciplinary relevance of this work and its resonance beyond the security community. |

#### Key output 6

Open Access: Yes

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reference:</b>          | Markatou, Evangelia Anna, Francesca Falzon, Zachary Espiritu, and Roberto Tamassia. "Attacks on encrypted response-hiding range search schemes in multiple dimensions." Proceedings on Privacy Enhancing Technologies (2023).                                                                                                                                                                                                                  |
| <b>URL:</b>                | <a href="https://doi.org/10.56553/popets-2023-0106">https://doi.org/10.56553/popets-2023-0106</a>                                                                                                                                                                                                                                                                                                                                              |
| <b>Type:</b>               | Conference paper                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Quality indicators:</b> | 1) Academic interest, other<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                 |
| <b>Motivation:</b>         | This paper shows that even advanced response-hiding range search schemes are vulnerable to information leakage. I was first co-author, leading the cryptographic analysis and guiding the implementation. This result underscored the risks of deploying encrypted databases without rigorous evaluation, showing that even advanced schemes can fail in practice. The paper was published in PoPETs, an important venue for privacy research. |

#### Key output 7

Open Access: Yes

|                   |                                                                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reference:</b> | Altman, Micah and Cohen, Aloni and Falzon, Francesca and Markatou, Evangelia Anna (Lilika) and Nissim, Kobbi and Reymond, Michel Jose and Saraogi, Sidhant and Wood, Alexandra, A principled approach to defining anonymization (May 9, 2022) |
| <b>URL:</b>       | <a href="https://dx.doi.org/10.2139/ssrn.4104748">https://dx.doi.org/10.2139/ssrn.4104748</a>                                                                                                                                                 |
| <b>Type:</b>      | Article, non-refereed                                                                                                                                                                                                                         |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality indicators: | 1) Academic collaboration and/or interdisciplinary engagement<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Motivation:         | This interdisciplinary paper, co-authored with researchers from MIT, Harvard, ETH Zurich, and others, bridged computer science and law. We developed a principled framework for defining anonymization under EU data protection. I translated formal privacy concepts into terms usable in legal and regulatory contexts. This project extended my research beyond my PhD topic, showing my ability to apply cryptographic expertise in a broader interdisciplinary and international setting. It demonstrates both my capacity to contribute independently to collaborations outside my immediate group and my commitment to research with direct societal relevance. |

#### Key output 8

Open Access: No

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:          | F+Cube (Future Female+ Faculty) Mentorship Program (2024–2025)                                                                                                                                                                                                                                                                                                                                                                   |
| URL:                | <a href="https://www.tudelft.nl/ewi/over-de-faculteit/afdelingen/software-technology/f-cube">https://www.tudelft.nl/ewi/over-de-faculteit/afdelingen/software-technology/f-cube</a>                                                                                                                                                                                                                                              |
| Type:               | Outreach/public engagement/advocacy, other                                                                                                                                                                                                                                                                                                                                                                                       |
| Quality indicators: | 1) Contribution to (academic) educational programme(s)<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                        |
| Motivation:         | I actively contribute to initiatives that broaden participation in computer science. In 2024–2025, I joined the F+Cube mentorship program, supporting early-career researchers in navigating the academic job market. This complements my other mentoring in programs such as the Brown Application Feedback initiative and the PETs Symposium mentorship event and reflects my long-term commitment to diversity and inclusion. |

#### Key output 9

Open Access: No

|                     |                                                                                                                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:          | Program Committee Member, ACM CCS 2025                                                                                                                                                                                                                                                                                                      |
| URL:                | <a href="https://www.sigsac.org/ccs/CCS2025/program-committee/">https://www.sigsac.org/ccs/CCS2025/program-committee/</a>                                                                                                                                                                                                                   |
| Type:               | Other, please describe:                                                                                                                                                                                                                                                                                                                     |
| Quality indicators: | 1) Academic collaboration and/or interdisciplinary engagement<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                            |
| Motivation:         | I serve on the Program Committee of ACM CCS 2025, one of the “Big Four” international computer security conferences. I review submissions and contribute to maintaining the high scientific standards of the field. This role reflects peer recognition of my expertise and my active engagement with the international research community. |

#### Key output 10

Open Access: Yes

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference:          | Halldórsson, Magnús M., Stephan Holzer, Evangelia Anna Markatou, and Nancy Lynch. "Leader election in SINR model with arbitrary power control." Theoretical Computer Science 811 (2020): 21-28.                                                                                                                                                                                                                                                                                                                                         |
| URL:                | <a href="https://doi.org/10.1016/j.tcs.2019.01.024">https://doi.org/10.1016/j.tcs.2019.01.024</a>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Type:               | Article, refereed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Quality indicators: | 1) Academic awards, prizes and/or grants directly related to this output<br>2) Optional: choose a second indicator<br>3) Optional: choose a third indicator                                                                                                                                                                                                                                                                                                                                                                             |
| Motivation:         | For my Master's thesis, I studied the classic distributed algorithms problem of leader election. I showed that by varying transmission power in a wireless network, leader election could be achieved in just two communication rounds, improving on prior logarithmic solutions. I led the research in this paper, contributing the key technical insight and formal analysis. The work won the Best Paper Award at SIROCCO 2017 and was later published in Theoretical Computer Science, showcasing my strong theoretical foundation. |

|                                             |     |
|---------------------------------------------|-----|
| Word count 2b. Key output (max. 700 words): | 700 |
|---------------------------------------------|-----|

### 3. Research idea

|                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title:                                                                                                           | Towards Secure and Efficient Databases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Key words (max. 5):                                                                                              | COMPUTER SECURITY, DATABASE SECURITY, PRIVACY ENHANCING TECHNOLOGIES, APPLIED CRYPTOGRAPHY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Research idea<br>(Domains <b>AES</b> and <b>SSH</b> : max 100 words.<br>Domain <b>Science</b> : max. 500 words): | <p>One of the greatest challenges in cryptography is <b>developing cryptographic algorithms that allow computation directly on encrypted data</b>. Such algorithms are invaluable where computation is performed on untrusted platforms, e.g. cloud computing. Although techniques such as Fully Homomorphic Encryption and Oblivious RAM provide ideal cryptographic guarantees, they remain impractical due to their high computational cost. In pursuit of more practical solutions, <b>Searchable Encryption (SE)</b> was developed. SE enables efficient queries over encrypted databases with such low computational cost that it has reached industry adoption, with MongoDB, industry leader among NoSQL databases, releasing <b>Queryable Encryption</b>.</p> <p>Unfortunately, the efficiency of SE comes at a price: information leakage. While the data and queries remain encrypted, SE schemes may reveal patterns such as which memory locations are accessed or whether a query has been issued before. What appears to be harmless leakage raises the <i>foundational</i> question in SE: just how dangerous is it in practice?</p> <p>Research has shown that <b>this leakage can indeed be exploited</b>. In 2012, the first reconstruction attack demonstrated that adversaries could infer user queries. Leakage from range queries was shown in 2016 to enable reconstruction of encrypted data - that is, an adversary could find the original (unencrypted) data. Since then, numerous leakage-abuse attacks have appeared, aiming to clarify the risks. Despite steady progress, the complexity of leakage and its implications remains insufficiently understood. Most attacks make unrealistic assumptions, such as adversaries with significant leakage or consider simplified databases.</p> <p>To advance the field, we must design practical attacks under more realistic assumptions. Only then can we assess the <b>true risks</b> of leakage in deployed systems. Insights from such attacks also inform mitigation techniques.</p> <p><i>One must not inquire merely whether encrypted databases leak information; rather, one must confront the <b>foundational</b> question: what can adversaries accomplish with such leakage?</i></p> <p>For absent an answer to this, no meaningful evaluation of security is possible.</p> <p>I will approach the problem from two angles: attacks and defenses. Most attacks have focused on single- or two-attribute databases. On the contrary, my work will focus on multi-attribute databases, which are more widely used. I will move beyond passive eavesdroppers to active adversaries who observe leakage and manipulate data, for example, by adding a fabricated patient record to a hospital database or sending well-crafted emails to an encrypted mailbox. Crucially, my attacks will use minimal leakage, reflecting more realistic adversarial capabilities.</p> <p>On the defense side, the reconstruction space, the set of all databases consistent with a given leakage, will play a central role. It can measure attack strength and serve as a foundation for designing mitigation techniques. In practice, this could mean the adversary cannot improve beyond guessing among 10,000 potential databases. I will systematically exploit this uncertainty to protect privacy while maintaining efficiency.</p> <p>This project not only tackles a foundational challenge in cryptography: by advancing this theory and translating it into practical defenses, it will enable secure, efficient encrypted databases, guide technologies already adopted in industry, support regulatory compliance (e.g., GDPR), and strengthen societal trust in cloud-based systems.</p> |

|                                 |     |
|---------------------------------|-----|
| Word count 3.<br>Research idea: | 500 |
|---------------------------------|-----|

## 4. Administrative details

### 4a. Personal details

|                                   |                   |
|-----------------------------------|-------------------|
| Title(s), initial(s), surname(s): | Dr. E.A. Markatou |
|-----------------------------------|-------------------|

### 4b. Doctorate

|                                          |                                                            |
|------------------------------------------|------------------------------------------------------------|
| University/College of higher education:  | Brown University                                           |
| Date of PhD award (dd-mm-yy):            | 22-10-2023                                                 |
| Supervisor(s) (' <i>Promotor(en)</i> '): | Professor Roberto Tamassia                                 |
| Thesis title:                            | Exploring Searchable Encryption Leakage from Range Queries |

### 4c. Prospective host institution

|                   |                                |
|-------------------|--------------------------------|
| Host institution: | Delft University of Technology |
| Research group:   | Cybersecurity                  |

### 4d. Work experience since completing your (first) PhD

List appointments chronologically, with your current post in the bottom row.

| Past appointments                                      | Appointment type                             | Period<br>(date-date) | FTE | Institution                    |
|--------------------------------------------------------|----------------------------------------------|-----------------------|-----|--------------------------------|
| Assistant professor<br>(If other, please specify here) | Permanent<br>(If other, please specify here) | 15-9-2023 - Present   | Y   | Delft University of Technology |

### 4e. Net academic research time

Calculate your net academic research time, as described in the explanatory notes. Do not include the calculation, only state the actual number of months.

|                                                                                                         |    |
|---------------------------------------------------------------------------------------------------------|----|
| Number of months:                                                                                       | 23 |
| If applicable: describe any special circumstances causing a reduction in productivity (max. 100 words): |    |
|                                                                                                         |    |

## Applicant statements

### Use of extension clause

If you use the extension clause, only provide the date of the confirmation email ([talent@nwo.nl](mailto:talent@nwo.nl)) granting the extension. You only need an extension if you exceed the year limit on the reference date.

Are you using the extension clause?: No

If yes, the extension was confirmed on: Choose the date

### By submitting this form, I declare that:

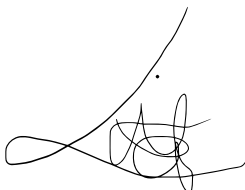
*By submitting this form, I declare that:*

- *I have completed this form truthfully and that I satisfy the nationally and internationally accepted standards for scientific conduct as laid down in the Netherlands Code of Conduct for Research Integrity 2018.*
- *I endorse the code of conduct for the use of laboratory animals and the code of conduct for biosecurity (in the event of dual use of the expected results) and will act accordingly, where applicable.*
- *The contact details in my ISAAC profile are up to date (i.e. email address postal address, phone number, personal details and correspondence language).*

Initial(s) and surname(s)<sup>1</sup>: E.A. Markatou

Place: Delft

Date: 12-9-2025



<sup>1</sup> To avoid gender effects, please do not state your first name.